Relationship Between the Pain Intensity at the Time of the Hamstring Muscle Injury and the Hyperintense T2 Weighted Volume of the Hamstring Muscle Lesion Measured by MRI

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Abstract

Objectives: The primary objective of this study was to analyze the correlation between the pain experienced at the time of a hamstring muscle injury and the hyperintense T2 weighted volume of the lesion measured on MRI. The secondary objectives were to analyze the differences in this pain with the lesion grade and the hamstring muscle head involved. **Design:** We performed a retrospective analysis of the data collected in a prospective, multicenter, noninterventional cohort study (HAMMER). **Patients:** A total of 61 patients were included in this analysis. **Interventions:** Patients with suspected hamstring muscle injury had an MRI after the injury. **Main outcomes measures:** They were evaluated for their pain intensity experienced at the time of injury with a Numeric Pain Rating Scale (NPRS) from 0 to 10. **Results:** There was a significant but weak correlation between the pain experienced at the time of hamstring muscle injury and the hyperintense T2 weighted volume ($r = 0.287$; $P = 0.025$). There were no significant differences in the pain according to the lesion grade ($F(3,57) = 0.982$; $P = 0.408$) nor according to the muscle involved ($F(3,57) = 1.013$; $P = 0.394$). **Conclusions:** Pain at the time of injury seemed to be weakly correlated with the extent of muscle lesion at medical imaging. Pain at the time of injury alone seems not enough to give us a clinical meaningful indication of the MRI hyperintense T2 weighted volume.

Key Words: hamstring muscle injury, edema, MRI, extent lesion, pain

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INTRODUCTION

Hamstring muscle injury is the most frequent injury in sports requiring sprints and acceleration such as soccer,[1,2] American football,[3] Australian football,[4] rugby,[5] hockey,[6] and track and field.[7,8] This noncontact injury leads to important physical, psychological, economic, and social consequences.[2,7,9–11] Improving its management thus represents an important challenge for all stakeholders around the athletes.

After a hamstring injury, main questions are about its consequences on performance, its risk of recurrence, and the time needed to return to sport. Responses to these questions are important to plan/anticipate the process of returning to sport in the best conditions. To do this, there is first a need to confirm the hamstring injury and its characteristics. This evaluation is of course first done clinically, but also often using medical imaging. In this process, the place of MRI is often debated.[12–17] Selecting patients for whom imaging adds value to the management seems to be a primary issue. The clinical examination could be of help, especially the evaluation of the pain at the time of injury.[18,19] Pain is the most reported symptom from the injured athlete.[20] It seems, therefore, of particular interest to explore whether pain intensity would be related to anatomical damage. This would help to determine whether the pain intensity at the time of injury would be a relevant information to help in the decision-making process of the MRI prescription after hamstring muscle injuries.

A muscle injury is the consequence of a stress or strain of the muscle tissue higher than its strength, which thus can lead to tissue damage.[21–23] This tissue damage can vary from edema to tissue rupture and can involve different structures of the muscle (myofibrils, tendons, and aponeuroses).[21] The destruction of intracellular structures (destruction of sarcomeres by the influx of calcium) and extracellular structures initiates the inflammatory response process and triggers the pain.[24] At the level of the muscles, there are mechanical or chemical (or

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The authors report no conflicts of interest.

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plurimodal) nociceptors. Mechanical nociceptors are activated by stretching of the fibers and their rupture: they are the first receptors activated and integrated at the cerebral level.[25] Chemical nociceptors are activated by the release of inflammatory substances such as bradykinins, prostaglandins, histamine, serotonin, and potassium ions.[26] They are particularly sensitive to concentrations of adenosine triphosphate (ATP) molecules (purinergic receptors) and to pH lowering (vanilloid receptors). We can, therefore, assume that the greater the muscle injury size, the more the ATP molecules and H^+ ions are released into the extracellular medium, consequently, the greater is the painful stimulus.[27] These pathophysiologic elements suggest that muscle tissue damage seems correlated with the pain stimulus. Besides the clinical aspect, MRI is a commonly used examination in routine. It has shown a high sensitivity and seems not to be operator dependent.[28] The T2 sequence allows the detection of tendinoligamentary and muscular lesions by highlighting the edematous and fluidic signal caused by muscular injuries. It seems quite logical to suggest that the greater the pain associated with the injury, the greater the size of the injury visible in T2 hypersignal on MRI. Such information would be of great interest for the clinicians to know whether they can use the clinical symptom of pain to estimate hamstring injury extent/size, and then can help to determine the interest of a medical imaging such as MRI.

In this context, the primary objective of this study was to analyze the correlation between the pain experienced at the time of a hamstring muscle injury and the hyperintense T2 weighted volume of the lesion measured on MRI. We hypothesized that the pain experienced by the patient at the time of injury is correlated with the MRI hyperintense T2 weighted volume. Secondary objectives were to analyze the differences in this pain with the lesion grade using the Peetrons classification or the hamstring muscle head involved (biceps femoris muscle long head (BFLH), biceps femoris muscle short head (BFSH), semitendinosus muscle (ST), and semimembranosus muscle (SM)).

METHODS

Experimental Protocol

We performed a retrospective analysis of data collected in the HAMMER (acronym for HAMstring MEchanics and mRi) study, a prospective, multicenter, noninterventional cohort study. The HAMMER study consisted in performing a thigh MRI in patients with a clinical suspicion of a hamstring muscle injury associated with a self-questionnaire completed by the patient describing the circumstances and mechanism of the injury. The HAMMER study received ethic approval from the Comité de Protection des Personnes d'Île-de-France V (CPP IDF 5: 17 059; N° ID-RBC; 2017-A03433-50). In this study, we used MRI data (extent and grade of injury, muscle(s) involved) with measurement of pain by the numerical pain rating scale (NPRS) from 0 to 10 experienced by the patient at the time of injury collected in the self-reported questionnaire.

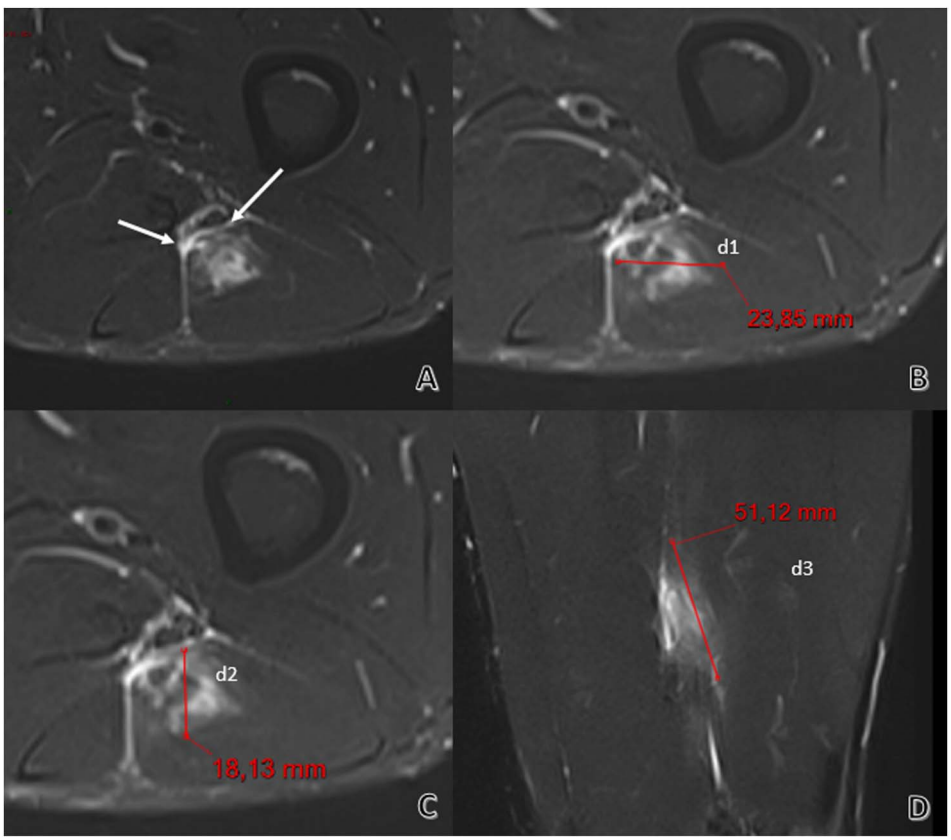

Figure 1. Examples of MR images centered on BFLH muscle tears, and measurements of the lesion. T2 fat saturated MR images in axial (A, B, and C) and coronal (D) views centered on BFLH muscle tears. The first measurement (d1) corresponds to the greatest axis of the lesion, and the second measurement corresponds to the axis perpendicular to the first (d2). The third measurement (d3), taken on the coronal section, corresponds to the maximum height according to the lesion. The measurement does not concern the effusion between the fascia (white arrow). From these 3 measurements, the lesion volume can easily be estimated.

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TABLE 1. Patients and MRI Characteristics

Physical Characteristics of the Population	Mean (SD) (Min–Max)
Age (yr)	27.5 (7.7) (18–47)
Height (cm)	181.6 (7.0) (171–206)
Weight (kg)	79.0 (10.5) (60–110)
BMI (kg.m^{-2})	24.0 (2.6) (19.6–31.7)
Demographic characteristics of the population	Numbers (%)
Sports	
Soccer	24 (39.3)
Track and field	20 (32.8)
Trail	5 (8.2)
Basketball	6 (9.8)
Other	6 (9.8)
Sport level	
Professional	17 (27.9)
Semiprofessional/national	17 (27.9)
Amateur	27 (44.2)
Medical history	
Previous hamstring muscle injury	39 (63.9)
Anterior cruciate ligament surgery	7 (11.5)
MRI characteristics of the lesions	Numbers (%)
Muscle injury	
No injury	6 (14.6)
One injury	44 (72.1)
More than 1 injury	11 (18.0)
Location	
Distal third	22 (36.1)
Middle third	34 (55.7)
Proximal third	19 (31.1)
Extensive (more than one third)	14 (23.0)
Muscle involved	
BFLH	38 (62.3)
BFSH	3 (4.9)
ST	15 (24.6)
SM	17 (27.9)
Peetrons grade	
Grade 0	6 (14.6)
Grade 1	14 (23.0)
Grade 2	37 (60.7)
Grade 3	4 (6.6)

Population

The patients were recruited in 2 radiology centers: the university hospital of Saint-Etienne and the mutualist clinic of Saint-Etienne. The patients were referred by their attending physician or by sports physician for MRI of the thigh after the suspicion of a hamstring muscle injury occurring during sports practice. The inclusion criteria were recreational/amateur, high-level, or professional athletes, aged 18 to 50 years, referred by their attending physician or their sports physician to a medical imaging facility for an MRI of the thigh in the context of a suspected acute hamstring muscle injury occurring during sports practice (physicians' diagnosis was made clinically (history and examination) without any recommendations), and dating back less than 21 days. Exclusion criteria were age less than 18 years and more than 50 years, and inability to understand the French language.

Data Collection

MRIs were performed on 1.5 or 3T clinical MR units. The axial and coronal sequences were performed using T1 and T2 weighted sequences with fat saturation. Parameters are available in **Supplemental Digital Content 1**, http://links.lww.com/JSM/A469. The musculotendinous insertion proximal to the ischial tuberosity, the insertions distal to the head of the fibula, and the proximal end of the tibia should be included in the field of view.

An acute hamstring injury was considered if high signal intensity was observed on T2 weighted MR images, as compared with that of the uninjured side, reflecting edema in or around the muscle. Images of chronic tendinopathy or chronic lesion (such as fibrosis) were not considered acute hamstring injuries. The extent of the MRI lesion was defined as any hyperintensity change at fluid-sensitive sequences of musculoaponeurotic and tendon structures (ie, hyperintense T2 weighted).[17] Signal changes between fascia were not considered as part of the lesion (ie, the effusion between the fascia, example given in Figure 1).[17] The size of the lesion was obtained using a prolate spheroid volume modeling, calculated from the 3 maximal dimensions of the lesion.[29] An example of an MRI of a hamstring muscle injury with measurement of the 3 major axes of the injury allowing calculation of the hyperintense T2 weighted volume is shown in Figure 1. The hamstring muscle head involved (BFLH, BFSH, ST, SM) was noted, and the "primary lesion" was defined as the muscle lesion with the largest hyperintense T2 weighted volume. Lesion grade was reported according to the Peetrons classification with 4 grades from 0 to 3.[30] All the data were collected by the same radiologist specializing in musculoskeletal imaging (SG).

At the end of the imaging examination, a self-reported questionnaire was given to the patient with questions about their physical characteristics (height, weight, age), their personal history of injury, their current sports practice, and the circumstances of the injury including the pain at the time of injury measured with the Numeric Pain Rating Scale (NPRS).

Statistical Analysis

Descriptive statistics was provided as count and percentage for categorical data and mean and SD, as well as a minimum and maximum, for the continuous variables. Correlation between the pain at the time of hamstring injury and the hyperintense T2 weighted volume was provided by using the Pearson correlation. The differences in this pain at the time of hamstring injury with the lesion grade using the Peetrons classification (0, 1, 2, 3) or the hamstring muscle head involved (BFLH, BFSH, ST, SM) were analyzed using a 1-way ANOVA. Statistical analyses were performed using JASP (JASP Team software, Version 0.14.1, University of Amsterdam, Netherlands). The level of statistical significance was set at $P < 0.05$.

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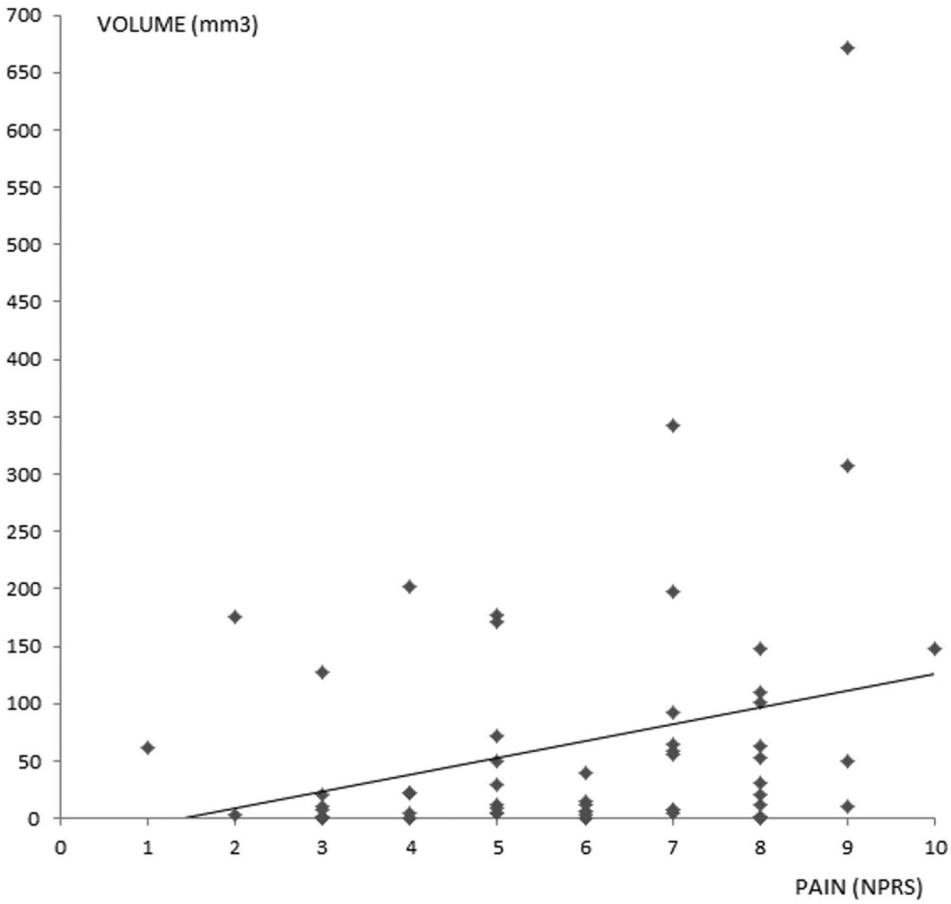

Figure 2. The correlation between the pain experienced at the time of a hamstring muscle injury measured by the Numeric Pain Rating Scale (NPRS) and the hyperintense T2 weighted volume of the lesion measured on MRI.

RESULTS

Population

From February 2018 to June 2021, we included 61 patients who were referred for hamstring MRI at the 2 radiology centers in Saint-Etienne and were included in the study. The main sports played were soccer (39%) and track and field (34%). Most patients had a previous hamstring muscle injury (61%). The physical and demographic characteristics of the population are presented in Table 1.

Hamstring Muscle Injuries

The mean delay between the day of injury and MRI was 7.8 ± 5.7 days; 79% were performed less than or equal to 10 days after injury. Most patients had a single lesion (81%). Most of the primary lesions affected the BFLH (61%), were located in the central part of the muscles (46%), and were grade 2 according to the Peetrons classification (60%). The main MRI characteristics of the hamstring injuries are represented in Table 1.

Analysis of the Pain Intensity, the Lesion Grade, and the Hamstring Muscle Involved, with the Hyperintense T2 Weighted Volume

There was a weak positive correlation between the pain at the time of injury and the MRI hyperintense T2 weighted volume ($P < 0.05$ R = 0.287) (Figure 2). There were no significant differences in the pain according to the lesion grade ($F(3,57) = 0.982$; $P = 0.408$) (Figure 3) nor according to the muscle involved ($F(3,57) = 1.013$; $P = 0.394$) (Figure 4).

DISCUSSION

The main findings of this study were that (1) there was a significant but weak correlation between the pain intensity at the time of injury and the MRI hyperintense T2 weighted volume and (2) there was no significant differences between this pain and the lesion grade or the affected muscle.

Our results reported that the pain at the time of injury seemed to be weakly correlated with the extent of muscle lesion at medical imaging. This weak correlation is consistent with pathophysiologic considerations: the greater the reactional edema, the more the pain is triggered.[21–26] However, although the correlation was significant, our results indicated that only 8% (r^2) of the variance in extent of edema on MRI was explained by the pain experienced at the time of injury. Pain at the time of injury alone seems not enough to give us a clinical meaningful indication of the MRI hyperintense T2 weighted volume. This information could be of help in clinical practice for determining the need of imaging and to manage injury. Indeed, if we need to know the extent of the MRI hyperintense T2 weighted volume, an MRI is needed.

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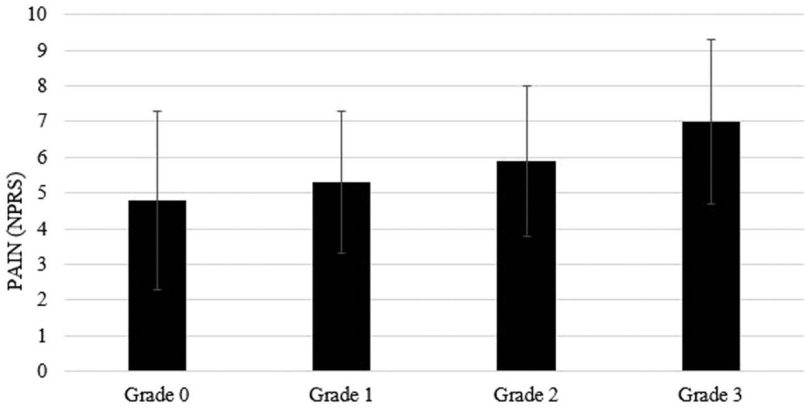

Figure 3. The Peetrons grade according to the pain at the time of hamstring injury measured using a Numeric Pain Rating Scale (NPRS) (mean ± SD).

The potential practical clinical applications of our present results could be that the pain at the time of injury can be of help to orient the decision making for MRI. For patients with minimal initial pain, we can expect a small MRI hyperintense T2 weighted volume. In such case, MRI could be of interest to confirm the fact that there is no tissue damage, and thus not to treat abusively the patient. Or at the inverse, to confirm the tissue damage and not to neglect the pain experienced by the patient. Conversely, in case of substantial pain, we can expect to see a more extensive MRI hyperintense T2 weighted volume. In these cases, MRI would seem to be desirable to carry out an assessment of the severity of the injury, in particular, to look for a hematoma or even a tendon rupture, the management of which is different and may lead to a discussion of a puncture or a surgical decision, respectively.[31] However, given our results reporting a weak correlation, it is important to note that if we need to know the extent of the MRI hyperintense T2 weighted volume, an MRI is needed.

As strengths of this study, we can report that the characteristic of patients included in this study seem to be consistent with the literature,[32] notably the predominance of isolated damage to a single muscle, predominance of BFLH and grade 2 lesion according to the Peetrons classification. Furthermore, this study is 1 of the few to analyze an anatomical–clinical relationship between a simple clinical examination parameter (pain at the time of injury) and an MRI parameter (injury extent) in hamstring injuries.

Some limitations of our study have to be acknowledged. We can consider our patient sample to be small compared with some cohorts in the literature.[10,33] In our study, the pain assessment was performed retrospectively, which may lead to recall bias, but the short delay after MRI (7.8 ± 5.7 days, ranging from 1 to 24) could have limited this bias. Furthermore, pain is subjective data whose evaluation remains variable between different individuals according to cultural factors, fatigue, and the emotional and psychological state in which the patient is.[34,35] More, our study evaluates only 1 clinical examination parameter (ie, the pain). Other clinical examination parameters should be explored in future studies (eg, pain during hamstring mobilization maneuvers, flexibility, strength). In addition, the combination of the initial pain and other clinical criteria could allow better optimization of the selection of patients eligible or not for MRI.

CONCLUSIONS

Our results reported a weak correlation between a simple clinical examination parameter (the pain at the time of injury) and an MRI parameter (the MRI hyperintense T2 weighted

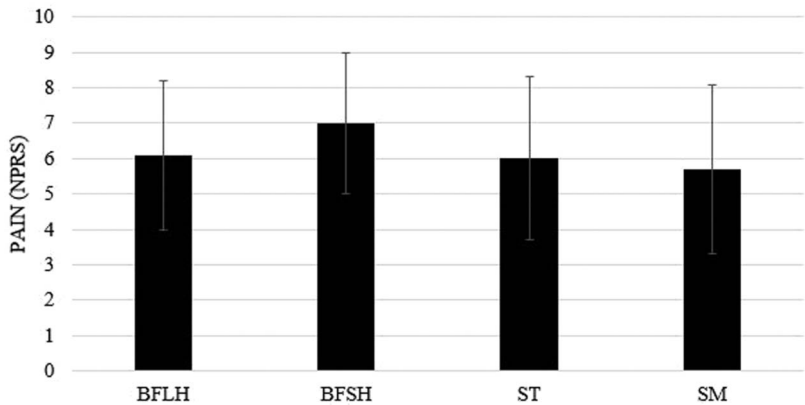

Figure 4. The hamstring muscle head involved according to the pain at the time of hamstring injury measured using an NPRS (mean ± SD). MRI parameters: T2 weighted sequences with fat saturation.

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volume). This is consistent with an anatomical–clinical relationship, however, pain at the time of injury alone only explained 8% (r^2) of the variance in extent of edema on MRI, and thus seems not enough to give us a clinical meaningful indication of the MRI hyperintense T2 weighted volume. This information could be of help in clinical practice for determining the need of imaging and to manage injury. Indeed, if we need to know the extent of the MRI hyperintense T2 weighted volume, an MRI is needed.

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